MovementLogger — Sensor & Module Catalog

9 Teile · 2026-05-21 09:26 UTC · BOM aus movement_logger_firmware · via crawl2pump

9 neu (7 Tage) 0 aktualisiert

Dev Boards 4

STEVAL-MKBOXPRO (SensorTile.box PRO Rev_C) OSS firmware USB-C pluggable

☑ USB-C ☐ WiFi ☑ Bluetooth ☐ GPS ☑ Motion sensors ☑ SD-card L×B×H 6.3 × 4 × 2 cm · 94 g

MCU: STM32U585 · Cortex-M33 @160 MHz · 2 MB flash / 786 KB SRAM · TrustZone · ultra-low-power

Datasheet: https://www.st.com/resource/en/user_manual/um3133-sensortilebox-pro-stmicroelectronics.pdf

OSS firmware: https://github.com/zdavatz/movement_logger_firmware

neu **STEVAL-MKBOXPRO (SensorTile.box PRO Rev_C) · STEVAL-MKBOXPRO @ DigiKey** **CHF 94.56**

STMicroelectronics @ DigiKey · in stock

Primary target board. USB-C for charge + STM32 DFU flashing. · USB-C · firmware: open-source

https://www.digikey.com/en/products/detail/stmicroelectronics/STEVAL-MKBOXPRO/19721662

GPS / GNSS 15

ARM Cortex JTAG/SWD 14-pin Cable (1.27 mm → 0.1" DuPont, ~100 mm) OSS firmware SMA / U.FL

☐ USB-C ☐ WiFi ☐ Bluetooth ☐ GPS ☐ Motion sensors ☐ SD-card L×B×H 10 × 0.5 × 0.2 cm · 5 g

no photo
ARM Cortex JTAG/SWD 14-pin Cable (1.27 m

neu **ARM Cortex JTAG/SWD 14-pin Cable (1.27 mm → 0.1" DuPont, ~100 mm) · Generic (Amazon / AliExpress OEMs) @ Generic (Amazon / AliExpress OEMs) (vendor page (reference))** —

Generic (Amazon / AliExpress OEMs) (manufacturer store)

Solderless JP2 plug. One end: 14-pin 1.27 mm shrouded female socket (mates with the STEVAL's FTSH-107 programming header — keyed, only goes on one way). Other end: 14 loose female DuPont leads labelled by SWD/JTAG net name. Pick the four…

https://www.amazon.de/s?k=JTAG+SWD+14+pin+1.27mm+cable+female+dupont

GPS/GNSS Magnetic Mount Antenna 3 m (active, SMA) OSS firmware SMA / U.FL

☐ USB-C ☐ WiFi ☐ Bluetooth ☐ GPS ☐ Motion sensors ☐ SD-card L×B×H 4 × 4 × 1.3 cm · 80 g

Datasheet: https://www.sparkfun.com/products/14986

neu **GPS/GNSS Magnetic Mount Antenna 3 m (active, SMA) · GPS-14986 @ DigiKey** **CHF 12.85**

SparkFun @ DigiKey · out of stock

Active GPS antenna (~28 dB LNA, 3 m RG-174 coax, SMA male). Magnetic base — sits on the board's non-metal top or the case lid for best sky view. Needs a U.FL → SMA pigtail to mate with the MAX-M10S breakout's U.FL connector — see next…

https://www.digikey.com/en/products/detail/sparkfun-electronics/14986/9682235

Interface Cable SMA → U.FL, 100 mm (pigtail)

OSS firmware · SMA / U.FL

☐ USB-C · ☐ WiFi · ☐ Bluetooth · ☐ GPS · ☐ Motion sensors · ☐ SD-card · L×B×H 10 × 0.3 × 0.3 cm · 3 g

Datasheet: https://www.sparkfun.com/products/9145

neu **Interface Cable SMA → U.FL, 100 mm (pigtail) · SparkFun 9145 @ sparkfun.com** — **USD 5.75**

SparkFun

100 mm pigtail: U.FL female (plugs into the SparkFun MAX-M10S breakout) ↔ SMA female bulkhead (mates with the antenna above's SMA male). Buy with the magnetic antenna or skip both if the onboard chip antenna is enough. · SMA / U.FL ·…

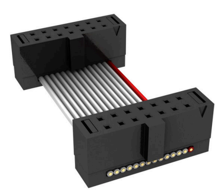
https://www.sparkfun.com/products/9145

Samtec FFSD-07-D-04.00-01-N — 14-pin 1.27 mm IDC ribbon (100 mm)

OSS firmware · SMA / U.FL

☐ USB-C · ☐ WiFi · ☐ Bluetooth · ☐ GPS · ☐ Motion sensors · ☐ SD-card · L×B×H 10 × 0.5 × 0.2 cm · 3 g

Datasheet: https://suddendocs.samtec.com/catalog_english/ffsd.pdf



neu **Samtec FFSD-07-D-04.00-01-N — 14-pin 1.27 mm IDC ribbon (100 mm) · FFSD-07-D-04.00-01-N @ DigiKey** — **CHF 9.21**

Samtec @ DigiKey · in stock

Solderless JP2 plug — 2×7 1.27 mm shrouded socket on both ends, mates with the STEVAL's FTSH-107 programming header. To break the ribbon's other end out to 0.1" DuPont (which slides onto the SparkFun MAX-M10S breakout's UART pins) you also…

https://www.digikey.com/en/products/detail/samtec-inc/FFSD-07-D-04-00-01-N/6613350

SparkFun GPS Breakout - MAX-M10S (Qwiic)

OSS firmware · Qwiic / I²C

☐ USB-C · ☐ WiFi · ☐ Bluetooth · ☑ GPS · ☐ Motion sensors · ☐ SD-card · L×B×H 2.54 × 2.54 × 0.6 cm · 4 g

Datasheet: https://cdn.sparkfun.com/assets/7/5/9/a/a/MAX-M10S_DataSheet_UBX-20035208.pdf

OSS firmware: https://github.com/zdavatz/movement_logger_firmware

neu **SparkFun GPS Breakout - MAX-M10S (Qwiic) · GPS-18037 @ DigiKey** — **CHF 35.80**

SparkFun @ DigiKey · in stock

Recommended MAX-M10S carrier. Open-hardware breakout. · Qwiic / I²C · firmware: open-source

https://www.digikey.com/en/products/detail/sparkfun-electronics/18037/16719314

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u-blox MAX-M10S GNSS module

OSS firmware · UART pins

☐ USB-C · ☐ WiFi · ☐ Bluetooth · ☑ GPS · ☐ Motion sensors · ☐ SD-card · L×B×H 1.01 × 0.97 × 0.25 cm · 0.60 g

Datasheet: https://content.u-blox.com/sites/default/files/MAX-M10S_DataSheet_UBX-20035208.pdf

OSS firmware: https://github.com/zdavatz/movement_logger_firmware

neu **u-blox MAX-M10S GNSS module · MAX-M10S-00B @ DigiKey** — **CHF 10.38**

u-blox @ DigiKey · in stock

Wired to UART4 @ 38400. The firmware's GPS source. · UART pins · firmware: open-source

https://www.digikey.com/en/products/detail/u-blox/MAX-M10S-00B/15712906

Enclosure 6

Hammond 1554G2GYCL — IP66 PC enclosure, clear lid (120×90×60 mm) OSS firmware enclosure

☐ USB-C ☐ WiFi ☐ Bluetooth ☐ GPS ☐ Motion sensors ☐ SD-card L×B×H 12 × 9 × 6 cm · 154 g

Datasheet: https://www.hammfg.com/electronics/small-case/plastic/1554.pdf

neu **Hammond 1554G2GYCL — IP66 PC enclosure, clear lid (120×90×60 mm) · 1554G2GYCL @ DigiKey** **CHF 22.77**

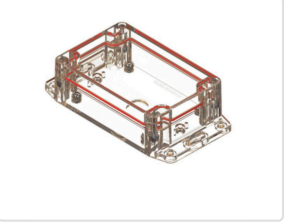
Hammond Manufacturing @ DigiKey · in stock

Polycarbonate IP66 project box, external 120 × 90 × 60 mm, clear PC lid. Lid sealed with O-ring gasket and four corner screws. RF-transparent so onboard GPS (and any radar) reads through the wall. The Hall sensor reads a passing magnet…

https://www.digikey.com/en/products/detail/hammond-manufacturing/1554G2GYCL/2359944

SERPAC RBF53P06C10C — IP67 PC enclosure, clear lid (120 × 80 × 40 mm) OSS firmware enclosure

☐ USB-C ☐ WiFi ☐ Bluetooth ☐ GPS ☐ Motion sensors ☐ SD-card L×B×H 12 × 8 × 4 cm · 75 g



neu **SERPAC RBF53P06C10C — IP67 PC enclosure, clear lid (120 × 80 × 40 mm) · RBF53P06C10C @ DigiKey** **CHF 21.97**

SERPAC @ DigiKey · in stock

Polycarbonate IP67 project box, external 120 × 80 × 40 mm (4.72 × 3.15 × 1.59 in), clear PC lid sealed with perimeter O-ring + stainless-steel screws into metal inserts. IP67 = submersion to 1 m for 30 min, the right call for a foilboard…

https://www.digikey.com/en/products/detail/serpac/RBF53P06C10C/17100309

Solderless alternative — bring-up only

If you don't want to touch the iron yet (or you want to verify the wiring before committing to a permanent build), the BOM carries two cable options that let you plug into JP2 without soldering. **Caveat:** a vibrating pumpfoil board can wiggle the cable loose mid-session — use this for bring-up / development, switch to the soldered rig below for production.

Reference videos (watch first)

- ControllersTech — GPS Module and STM32 (NEO-6M): generic STM32 + u-blox GPS over UART. NEO-6M but the wiring is identical to MAX-M10S (TX↔RX crossover, GND, 3.3 V). Watch this one if you watch one.
- SparkFun — MAX-M10S Product Showcase: the specific GPS breakout from this BOM (PRT-18037). Shows the U.FL connector and the UART header pins you'll connect to.
- Chris Park — STM32 + u-blox UART NMEA, **wiring from 11:33**: the closest-fit hookup demo. Uses M8N (same pinout as the MAX-M10S), shows 4 DuPont female leads plugged onto VCC/GND/TX/RX on the GPS module. Translate directly to the STEVAL: JP4 → 3 V & GND, JP2 → TX/RX (crossover: STEVAL TX → GPS RX, STEVAL RX → GPS TX).

There is no exact "STEVAL-MKBOXPRO + solderless JTAG cable + MAX-M10S" video — that combination is bespoke to this build. The two videos above cover the load-bearing concepts (UART wiring + the specific GPS module).

Pick a cable path

Path	Parts	Buyer experience
A · Distributor-grade	Samtec FFSD-07-D-04.00-01-N + 1.27→2.54 mm SWD adapter PCB + 4× DuPont jumpers + 2.54 mm 7-pin header	Real CHF prices, 3 SKUs to assemble (~CHF 15 total). Adapter PCB is the only AliExpress-only piece (~CHF 3, no canonical MPN — search "Cortex SWD 14-pin adapter").
B · Turnkey	Single 14-pin 1.27 mm → DuPont cable + 2.54 mm 7-pin header	One Amazon.de order (~CHF 8), card above shows the placeholder image since the category has no MPN. 2 SKUs.

Both paths need the same JP4 3.3 V power tap and the same 4 DuPont connections — the difference is only how you get from JP2's fine-pitch socket to standard 0.1″ DuPont leads.

Step 1 · JP4 power tap (3.3 V)

1. The JP4 connector is the 7-pin 2.54 mm strip on the side of the board (Multicomp MC-SVT1-S07-G).
2. **Set the JP4 domain switch to 3 V** (not 1.8 V).
3. Power on the box and **meter the JP4 3 V pin**: must read **3.25–3.40 V**. Power off again.
4. Slip a standard **2.54 mm 7-pin female header strip** onto JP4 (no solder — slides on). DuPont jumper from the 3 V pin → GPS VCC.

Step 2 · JP2 data lines (UART4)

1. Plug the JTAG cable into JP2 (FTSH-107 keyed/shrouded socket — only goes on one way).
2. Wire 3 DuPont jumpers from the cable's DuPont side (or, on path A, from the adapter PCB's 2.54 mm side) to the **SparkFun MAX-M10S breakout's UART header**:

STEVAL JP2 pin	Net (MCU)	→	SparkFun GPS pin
13	UART4_RX / PA1	→	**TX**
14	UART4_TX / PA0	→	**RX**
7 or 11	GND	→	**GND**

Note the TX/RX crossover (each end's "TX" pairs with the other's "RX"). The SparkFun breakout's UART header has male 2.54 mm pins populated — female DuPont leads slide on directly. **Do not touch SWD pins (4 / 6 / 8 / 10 / 12)** — they're the debug bus; bridging one means bricked debug access.

Step 3 · Bring-up test

1. Insert the SD card. Take the box **outdoors, clear sky**. Power on.
2. Cold fix is typically reached in **~30–60 s**. Confirm with a growing `GpsNNN.csv` on the SD card, or the BLE SensorStream `gps_valid` flag, or the antenna-test buzzer build.
3. No fix in a few minutes? **Swap pin 13 ↔ pin 14** — reversed TX/RX is by far the most common wiring mistake and swapping is harmless. Then re-check GND continuity and GPS VCC = 3.3 V under load.

When to switch to soldering: once bring-up works and you're ready for water trials. Vibration + spray + a loose cable will end a session early; the soldered rig below is the durable answer.

Soldering the u-blox MAX-M10S GPS to the STEVAL-MKBOXPRO

The MovementLogger firmware reads GPS over **UART4** on MCU pins **PA0 (TX) / PA1 (RX)** at **38400 baud, 8N1, 3.3 V logic**. The MAX-M10S is **not** on the box — it must be wired by hand to **JP2**, the 14-pin programming connector (Samtec FTSH-107-01-L-D, 1.27 mm pitch). Full source: GPS_SOLDERING.md.

Tools you need

- Fine-conical-tip iron, flux, magnification (10× loupe or microscope), tweezers, steady bench.
- **30 AWG** enamelled / wire-wrap wire (≤ 0.5 mm), four lengths.
- Multimeter (continuity + DC volts).
- Kapton tape + a dab of hot glue for strain relief.

JP2 pinout — pin 1 = square pad / silk triangle

Pin	Net	Use?
1	1V8 rail (JTAG ref)	**NOT** power — 1.8 V, weak
4 / 6 / 8 / 10 / 12	SWDIO / SWCLK / SWO / JTDI / NRST	Leave alone (debug bus)
7	GND	OK for GPS GND
11	GND	OK for GPS GND
13	**UART4_RX (PA1)**	**GPS TX → here**
14	**UART4_TX (PA0)**	**GPS RX → here**

Pins 3, 5, 9 carry no signal — leave unconnected. **Meter-verify** pin 1 against the silkscreen triangle and continuity-check every pin before soldering — connector orientation differs between board builds.

Wiring — note the TX/RX crossover

GPS pin	→	JP2 pin	Net (MCU)
TX (GPS out)	→	**13**	UART4_RX (PA1)
RX (GPS in)	→	**14**	UART4_TX (PA0)
GND	→	7 or 11 (meter-verified)	GND
VCC 3.3 V	→	**JP4 3 V pin** (not JP2!)	Board 3 V rail

GPS power — 3.3 V from JP4

- JP4 is the 7-pin extension-board connector (Multicomp MC-SVT1-S07-G). Set its domain switch to **3 V** (not 1.8 V).
- Power the box and **meter the JP4 3 V pin** — must read **3.25–3.40 V** before connecting GPS VCC. Power down again before soldering VCC.
- GPS GND and box GND must be common — take GND from JP4 or a JP2 GND pin.
- No graceful shutdown: the Hall-sensor magnet cuts the whole supply rail at once (firmware F-PWR-5). The on-SD format tolerates it.

Safety — read before the iron is hot

- **3.3 V logic only.** PA0/PA1 are **not 5 V tolerant.** A 5 V GPS breakout's TX into PA1 will destroy the MCU — level-shift or use a 3.3 V-native MAX-M10S carrier (e.g. the SparkFun breakout above).
- **Never bridge to the SWD pins** (4/6/8/10/12) — solder bridge → bricked debug access.
- Keep each pad heated **< 2 s**. FTSH-107 pads lift if cooked.
- **Strain-relieve all four wires** with Kapton + hot glue close to JP2 — a tugged wire rips the pad off the PCB.

Procedure

1. **Power down completely** — switch S1 off, USB unplugged, LiPo disconnected.
2. **Locate JP2** (14-pin fine-pitch header by the MCU). Identify pin 1.
3. **Continuity-check before soldering** (meter, board off): JP2 pin 13 ↔ MCU PA1, JP2 pin 14 ↔ MCU PA0, candidate GND pin ↔ battery (−). Confirm, don't assume.
4. **Pre-tin** the three JP2 pads and the wire ends with flux.
5. **Solder the data wires:** pin 13 ← GPS TX, pin 14 ← GPS RX. Then **GND**.
6. **GPS VCC last:** with the board powered, confirm 3.3 V at the JP4 3 V pin, power down again, solder the VCC wire.
7. **Inspect under magnification** for bridges — especially pin 14 to pin 12 and onto the SWD pins.
8. **Strain-relieve** all four wires (Kapton + hot glue).

Bring-up test

1. Insert the SD card, take the box **outdoors with clear sky**, power on.

2. The firmware sends the u-blox UBX config on boot, then expects 10 Hz NMEA. First fix is typically reached within **~30–60 s cold**.
3. Confirm a fix by **any** of: a growing `GpsNNN.csv` on the SD card · the BLE SensorStream `gps_valid` flag going set · (antenna-test build, `make GPS_ANTENNA_BEEP=1`) the buzzer beeping N times every 10 s with N = satellites.

No fix after several minutes outdoors? — in order of likelihood

1. **Swap the two data wires** (pin 13 ↔ pin 14). TX/RX reversed is the single most common wiring mistake and swapping is harmless to try.
2. Re-check **GND** continuity (cold/again) and that GPS VCC really is 3.3 V *at the module* under load.
3. Confirm the module is a 3.3 V part and actually powered (most MAX-M10S carriers have a power LED).
4. Give it a longer cold-start with an unobstructed sky view; indoors it will never fix.